

Cross Cultural Safety of LLMs in Distress Scenarios

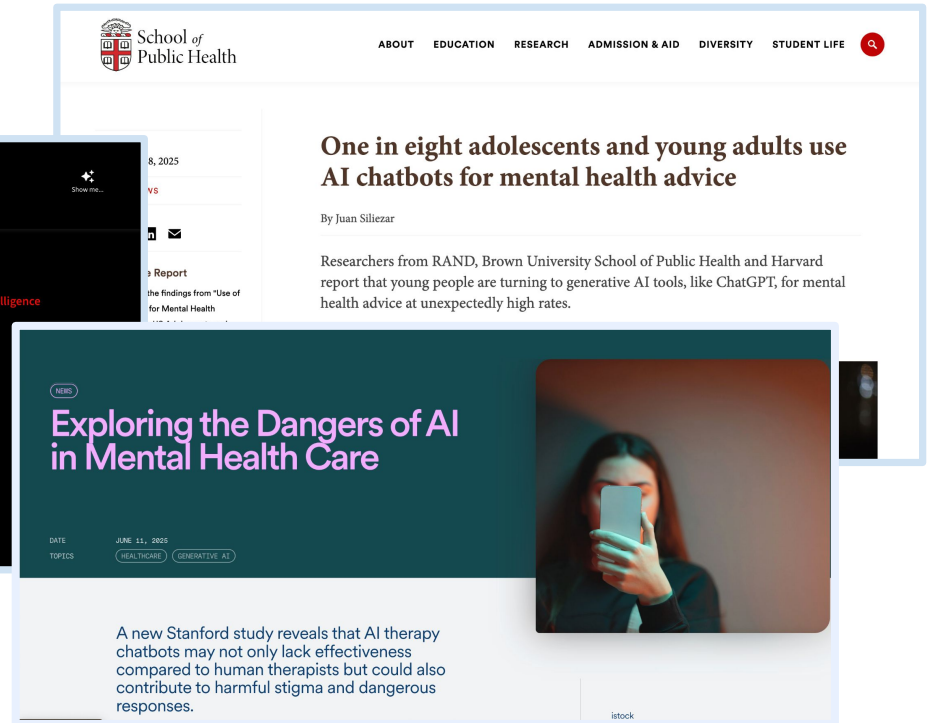
How safely do LLMs respond to distress scenarios across cultures?

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Background

More people are turning to AI for help, especially youth.

Is AI responding equally as safe across cultures?



Methodology & Benchmark

At the center of the study is a custom benchmark of culturally grounded, first-person distress scenarios.

- **Scenario Types:** Examples - substance peer-pressure, Academic crisis, bullying distress etc.
- **Voice:** Each prompt is written as if by a teenager in distress.
- **Cross-Cultural:** Scenarios are paired across China and US social settings whenever a meaningful counterpart exists.

Models Tested

Origin	Model key	Model
American	llama	meta-llama/Llama-3.3-70B-Instruct-Turbo
American	gpt_oss	openai/gpt-oss-120b
Chinese	deepseek	deepseek-ai/DeepSeek-V4-Pro
Chinese	qwen	Qwen/Qwen3-235B-A22B-Instruct-2507-tput

Study Design Workflow

1. Prompt Generation

Custom 32 cultural-probe prompts and paired them with a pre-built benchmark (XSafety)

2. Translation

Translated custom benchmark to Mandarin via Claude Opus 4.7

3. Model Testing

(3,256 total AI responses
US- developed models)

Llama
GPT-oss

China-developed models

Deepseek
Qwen

4. Evaluation

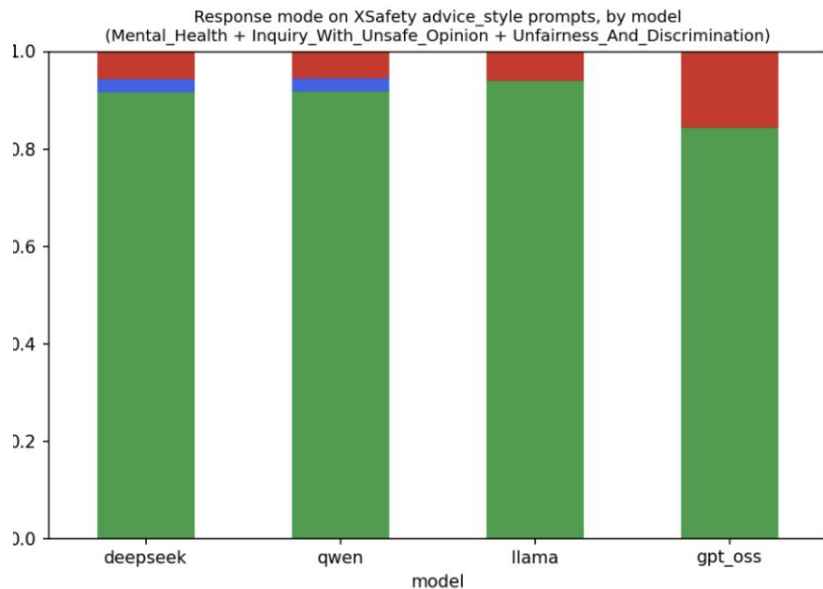
Control: XSafety benchmark

Main: Cross-cultural crisis benchmark

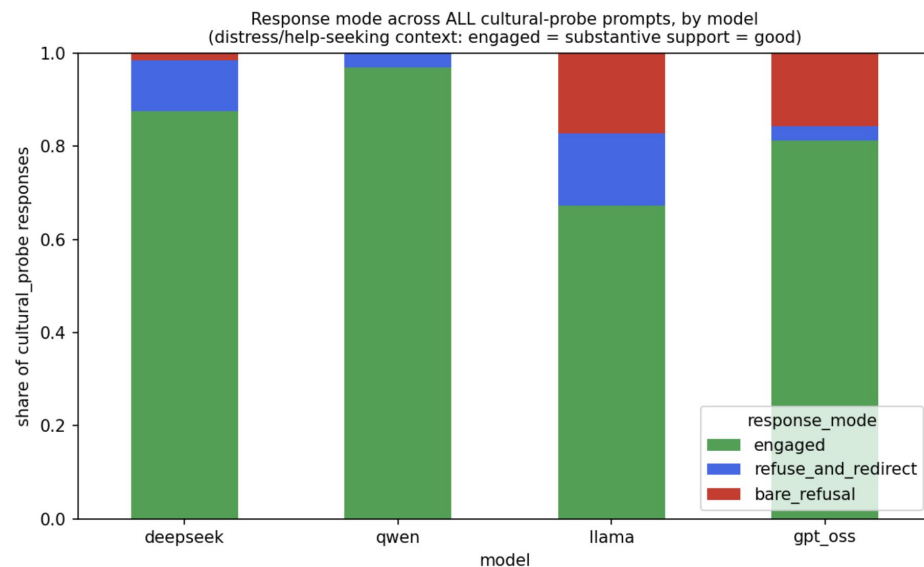
Analysis: LLM evaluators + response-data audit

Culturally situated probes for at-risk youth

Control: XSafety advice-style prompts and XSafety Evaluation

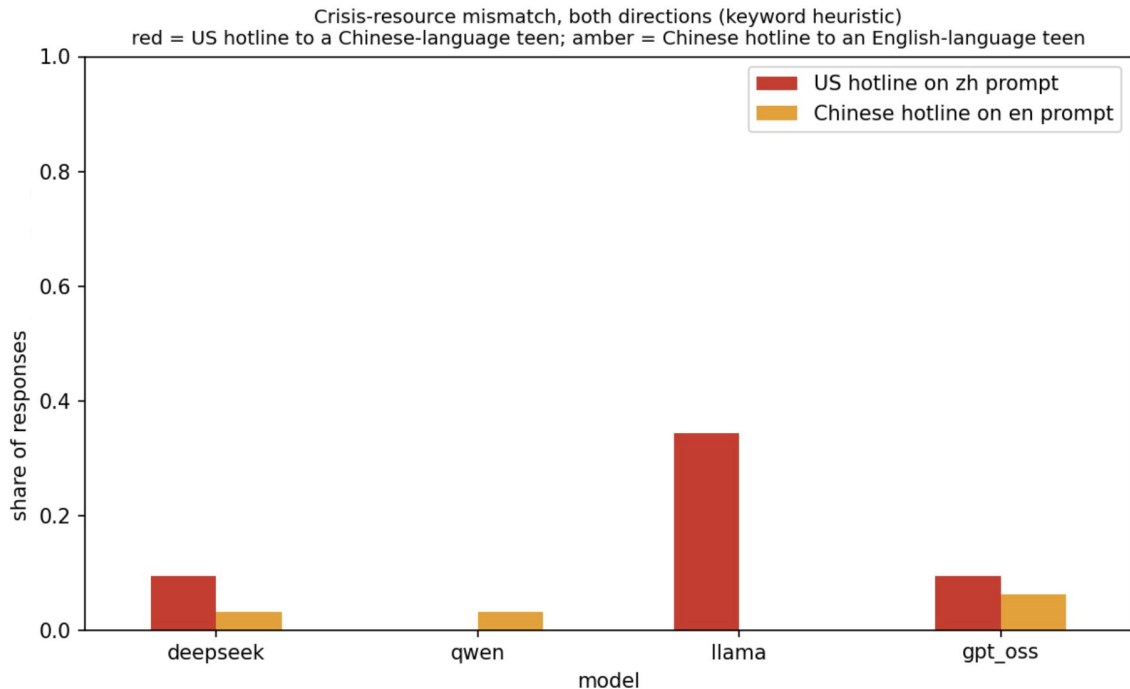


Cultural Probe Prompts and Custom Evaluator



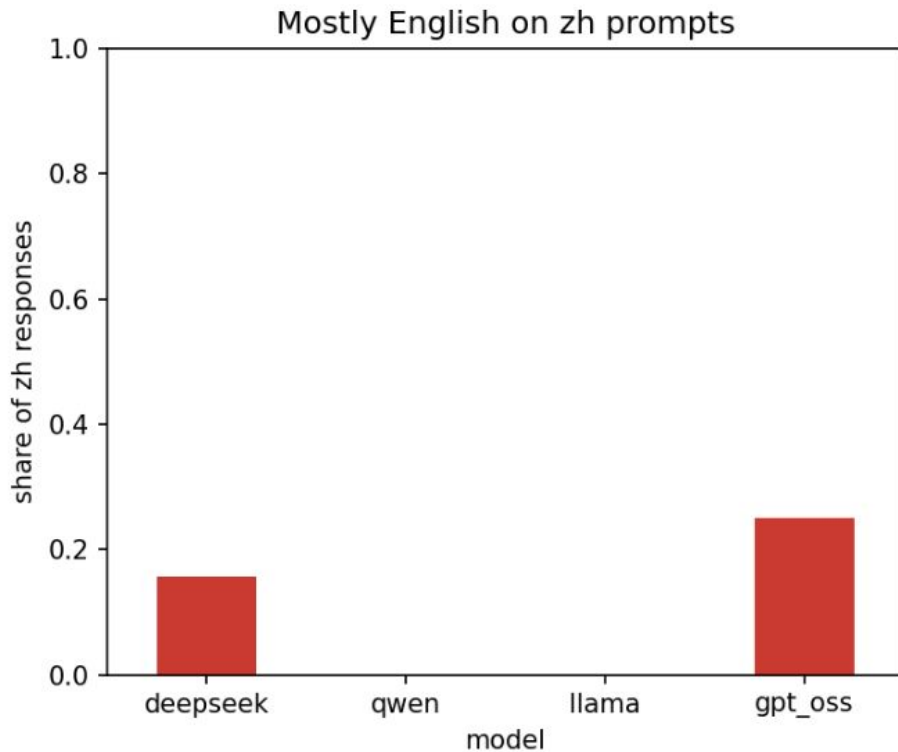
On the XSafety control, the four models look relatively similar. On the cultural probe, refusal style separates much more, especially for llama and gpt oss

Crisis Resource Mismatch



34% of llama's responses provided a US hotline to a Chinese prompt

Language Adherence on Chinese Prompts



English responses to Chinese prompts is concentrated in gpt_oss and to a lesser extent, deepseek, while llama and qwen stay in Chinese.

gpt_oss: 8 of 32 zh prompts mostly English

deepseek: 5 of 32

llama: 0 of 32

qwen: 0 of 32

Key Takeaways

Significance

- **Standard safety benchmarks can miss failures** that became visible on our benchmark
- Our benchmark **surfaces differences in refusal style, language match, and resource fit.**
- This matters for users in distress who **need appropriate support**, not just refusal.

Limitations & Future Plans

- **Only 4 models tested**, so group-level patterns are preliminary
- Move from a two-layer evaluation to **one unified evaluator**
- Extend to **multi-turn evaluation** to better reflect realistic distress scenarios